Jiawei Sun

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Summary

Ph.D. candidate in machine learning at Rensselaer Polytechnic Institute. Research spans diffusion models, vision-language models, and graph neural networks, with work on curriculum learning, feature purity, and structured sparsification. Developed PyTorch pipelines for multimodal pretraining, evaluation, and representation analysis on multi-GPU systems.

Education

Aug 2021 – Present **Rensselaer Polytechnic Institute**, Troy, NY
Ph.D (Expected) in Electrical, Computer, and Systems Engineering

Aug 2019 – Apr 2021 **University of Michigan**, Ann Arbor, MI
M.S. in Electrical and Computer Engineering

Aug 2015 – Jul 2019 **Northwestern Polytechnical University**, Xi'an, China
B.Eng. in Information Engineering

Publications

June 2026 **A Theoretical Analysis of Curriculum Training in Diffusion Models**

ICML 2026 Workshop on Foundations of Deep Generative Models (FoGen), 2026. Jiawei Sun, Shuai Zhang, Hongkang Li, Sijia Liu, Pin-Yu Chen, Meng Wang

Sep 2025 **Contrastive Learning with Data Misalignment: Feature Purity, Training Dynamics and Theoretical Generalization Guarantees**

Neural Information Processing Systems, 2025. Jiawei Sun, Shuai Zhang, Hongkang Li, Meng Wang

Jun 2025 **Theoretical Learning Performance of Graph Networks: the Impact of Jumping Connections and Layer-wise Sparsification**

Transactions on Machine Learning Research, 2025. Jiawei Sun, Hongkang Li, Meng Wang

Apr 2022 **Modelling learning in C. elegans chemosensory and locomotive circuitry for T-maze navigation**

European Journal of Neuroscience, 2022. Bennet G. Sakelaris, Zongyu Li, Jiawei Sun, Shurjo Banerjee, Victoria Booth, Eleni Gourgou

Research Experiences

Jan 2025 – Now **Diffusion Models with Noise Curriculum Learning** [code],
Advisor: Prof. Meng Wang

Pretraining dynamics and evaluation stability analysis for diffusion models

- Analyzed convergence behavior and feature learning dynamics of diffusion denoisers under noise-conditioned training, identifying regime-dependent evaluation variance across noise levels
- Developed a noise curriculum ordering samples from high-noise to low-noise regimes to stabilize early optimization and obtain more reliable pretraining signals for model comparison
- Introduced a sparsity curriculum that selectively masks parameter updates to prevent dominant-feature overfitting, improving representation robustness and reducing evaluation noise at low-noise regimes

Jan 2024 – Jan 2025 **Contrastive Learning with Data Misalignment in VLMs** [code],
Advisor: Prof. Meng Wang

Feature-learning evaluation for vision-language models

- Analyzed pretraining dynamics of contrastive learning under image-text misalignment, identifying how spurious cross-modal correlations inflate apparent performance without improving representation
- Formalized feature purity as an evaluation signal to distinguish genuine capability gains from noise-induced or artifact-driven improvements in base model representations
- Built a large-scale multimodal pretraining and evaluation pipeline combining noisy web data with synthetic recaptioning to study evaluation stability across data quality regimes
- Demonstrated that theory-guided sparsification by pruning low-purity neurons improves evaluation consistency and reduces inference cost while preserving representation clustering quality

Sep 2021 – Sep 2023 **GCNs with Skip-connections and Sparsification** [code],
Advisor: Prof. Meng Wang

NTK-guided convergence analysis and theory-driven design of scalable GCNs

- Developed an NTK-based analysis to characterize convergence behavior and optimization stability of GCN training under graph sparsification
- Derived theory-motivated conditions linking effective adjacency sparsity, depth, and representation mixing, explaining when aggressive sparsification preserves performance
- Designed skip-connection architectures that enable deeper layers to operate on more sparsified graphs while maintaining predictive accuracy
- Validated consistent improvements in training stability and efficiency on citation and OGB-style benchmarks across a range of sparsification levels

Sep 2019 – Apr 2020 **Image Processing for C. elegans Locomotion in Mazes** [code],
Advisor: Research Scientist Eleni Gourgou

Motion-trail extraction from microscopy video

- Extracted maze contours via Chan-Vese active contour and SVD, and aligned a T-shape polygon to the maze using the Procrustes Transformation
- Applied Frame Difference for trail extraction and K-Nearest Neighbor smoothing for trajectory refinement

Jan 2020 – Apr 2020 **Deep Neural Network for Spectrum Unfolding**,
Advisor: Prof. Alfred Hero

Proposed an RNN architecture mimicking projected gradient descent from optimization theory; implemented end-to-end in PyTorch.

Awards and Honors

Nov 2018 Honorable Mention, International Mathematical Contest in Modeling

Nov 2017 First Prize Scholarship, Northwestern Polytechnical University (Top 15% of 200 students)

Nov 2016 National Scholarship, Northwestern Polytechnical University (Top 0.2% nationwide)

Skills

Programming Languages: Matlab, Julia, Python, LaTeX, PyTorch